What is a *compound data structure* in Prolog?

In **Prolog**, a **compound data structure** (or **compound term**) is a way to **group multiple pieces of information together** under one name — like a record, object, or struct in other languages.

Definition

A **compound term** has:

```
Functor(Arg1, Arg2, Arg3, ...)
```

Where:

- **Functor** → the *name* of the structure (must start with a lowercase letter)
- **Arguments** → one or more data values (which can themselves be atoms, numbers, or other structures)

Examples

1. Simple facts

```
point(3, 4).
```

- Functor: point
- Arguments: 3, 4
- Meaning: a point at coordinates (3,4)

2. Nested (compound inside compound)

```
circle(point(3,4), 5).
```

- Functor: circle
- Arguments:
  - `point(3,4)` (itself a compound term)
  - 5 (the radius)

So this means:

A circle centered at (3,4) with radius 5.

Question:

Consider the following Prolog facts that represent details about different cars using a compound data structure:

```
car(toyota, model(corolla), engine(1600), color(blue, white, yellow)).
car(honda, model(civic), engine(1800), color(red)).
car(nissan, model(altima), engine(2000), color(black, grey)).
```

1. Write a Prolog query to find all cars that have **blue** as one of their colors.
2. Write a Prolog query to find the **engine capacity** of the **Toyota** car.
3. Explain how compound structures like `color(blue, white, yellow)` are useful in representing data in Prolog.